

(TAO Class)		
UNREP		CONTRACT NO. TBD
ITEM NO. PWS	CATEGORY "A"	
HPAC OVERHAUL & ELECTRICAL		

1.0 ABSTRACT

The purpose of this item is to perform the 18,000 hour maintenance on the ship's high pressure air compressors (HPACs) **AND** perform the electrical and condensate drain upgrade (due to obsolete equipment installed).

2.0 REFERENCES

- 2.1 NAVSEA Tech Manual T6220-AS-MMO-010, High Pressure Air Compressor Model: N20NL-11C
- 2.2 NAVSEA Tech Manual S6220-DR-OMI-B10 AIR COMPRESSORS HIGH PRESSURE OIL-FREE MODELS N20NL-11G AND N20NL-11H WITH PROGRAMMABLE LOGIC CONTROLLER

3.0 ITEM LOCATION/DESCRIPTION

3.1 Location/Quantity

3.1.1 Locations

- a. MSC to ship to Contractor's Facility

3.1.2 Quantity: Two (2)

3.2 Item Description/Manufacturer's Data

3.2.1 Manufacturer: Dresser-Rand Company Model Number: N20NL-11C

3.2.2 Serial number for compressor 1 is XM51224.

3.2.3 Serial number for compressor 2 is XM51318.

4.0 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL/SERVICE: None

5.0 NOTES:

- 5.1 Do not allow grease or oil to contaminate parts that contact the air stream. Ensure all lockout/tagout procedures are in accordance with

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current shipboard instructions.

5.2 Refer to the manufacturer's technical manual for detailed, step-by-step procedures for the following maintenance action.

6.0 QUALITY ASSURANCE REQUIREMENTS: None additional

7.0 STATEMENT OF WORK REQUIRED

7.1 Two (2) used HPAC Compressors as identified within Para 3.0, currently located in Alabama Shipyard will be delivered to the OEM approved facility for shop overhaul.

7.2 Drain and dispose of any liquid in accordance with all federal, state, and local regulations.

7.3 Temporarily remove crossheads and connecting rods from the compressor.

7.3.1 Remove and replace connecting rod crankpin bearing shells. When installing new shells, ensure that the oil passages are currently aligned.

7.3.2 Clean all crosshead and connecting rod parts with an approved solvent, paying particular attention to all oil passages.

7.3.3 Inspect and measure all critical dimensions of the crosshead, crosshead pin, and retaining ring. Take a gage reading of the connecting rod crankpin bearing. Replace any components exceeding allowable tolerances or are otherwise damaged.

7.4 Remove the crankshaft and main bearings.

7.4.1 Clean the crankshaft, paying particular attention to the journals surfaces and or passages.

7.4.2 Inspect the crankshaft for nicks, scratches other surface defects or cracks. Perform a magnetic particle test.

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- 7.4.3 Measure the critical dimensions of the crankshaft. Submit an "as found" condition report including recommendation to replace the crankshaft if allowable tolerances are exceeded.
- 7.4.4 Check the main bearing saddles for irregularities or debris.
- 7.4.5 Replace all main bearings and take gage readings to ensure correct clearances.
- 7.4.6 Reinstall crankshaft.
- 7.5 Thoroughly clean the frame and oil sump.
 - 7.5.1 Drain lube oil sump.
 - 7.5.2 Remove all dirt or sludge deposits.
 - 7.5.3 Replace O-ring in sump drain plug.
 - 7.5.4 Clean all oil pipes, including those previously removed from compressor and the lube oil strainer.
 - 7.5.5 Inspect all portions of the frame and sump for damage or other defects.
- 7.6 Disassemble and inspect the oil pump. Pay particular attention to the rotor assembly. Submit an "as found" condition report. Reassemble the oil pump.
- 7.7 Replace the crankshaft oil seal.
- 7.8 Clean and inspect the piston rod oil wiper assemblies.

Note: Before disassembling a wiper ring matchmark the ring segments to permit correct reassembly.

- 7.8.1 Examine the wiper rings for cracks or breakage. If one segment needs replacement, change the entire set.

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- 7.8.2 Clean the stuffing boxes before reinstalling rings. Reinstall rings.
 - 7.9 Temporarily remove and overhaul the freshwater pump.
 - 7.9.1 Disassemble freshwater pump and submit an "as found" condition report.
 - 7.9.2 Reassemble and reinstall freshwater pump.
 - 7.10 Clean the cooling water system.
 - 7.10.1 Open and inspect the air coolers and cooling water heat exchanger. Submit an "as found" condition report.
 - 7.10.2 Clean the cooling water system of dirt and debris. MSCREP to inspect.
 - 7.10.3 Close the cooler with new gaskets.
 - 7.11 Hydrostatically test the cooling water system.
 - 7.12 Refurbish the manual condensate drain manifold.
 - 7.13 Refurbish the drain line check valves.
 - 7.14 Refurbish the unloading valve and unloading valve pilot valve.
 - 7.15 Perform system upgrade to the drains (as are obsolete and not supported) "Electrical and Condensate Drain system related" to allow for compressors to have supportability for future repairs/service requirements.
 - 7.16 Prepare equipment for testing and accomplish operational test in OEM Workshop in the presence of MSCREP.
 - 7.17 After completion of all portions of this contract, coordinate arrange for transportation of Government Materials, two (2) Overhauled Condition Code "A" HPAC Compressors as identified within Para 3.0, from Contractors OEM Workshop back to LP-26 MSC

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Norfolk Warehouse.

7.18 Reports

- 7.18.1 When inspection, service and testing reveal any deficiency a condition report is to be submitted to the MSCREP with recommended repairs and parts required. Submit one (1) electronic copy in Portable Document Format (PDF).
- 7.18.2 Submit a report that includes "as released" condition of the HPACs. Submit one (1) electronic copy in Portable Document Format (PDF).
- 7.18.3 All reports and checklists are to be completed and signed by the person who carried out the inspection and maintenance work and countersigned by the Company's representative.

7.19 Painting/Lagging

- 7.19.1 Prepare, prime, paint and restore all new, soiled, and disturbed surfaces to match surrounding surfaces.

7.20 Manufacturer's Representative

- 7.20.1 Provide the services of an OEM authorized field service representative to accomplish all work and testing on the HPACs identified in paragraph 3.0 ensuring compliance with the manufacturer's standards and performance specifications.

- 7.20.2 Possible Source

RESERVED

- 7.21 Preparation of Drawings: Provide Tech Manual/Drawings (all relevant data) for the Electrical and Condensate Drain system related upgrades performed.

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8.0 GENERAL REQUIREMENTS: None additional